

CATEMS: A Lifecycle Model of Compromised AI Teammate Attacks, Effects, and Mitigation Strategies for Enhanced Resilience

Proceedings of the Human Factors and
Ergonomics Society Annual Meeting
1–7

© Human Factors and
Ergonomics Society 2026

Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/10711813261475221
journals.sagepub.com/home/pro



Sarah V. Mendoza¹, Yayun Tian¹, Beau G. Schelble¹,
Allyson Hauptman², and Lionel P. Robert³

Abstract

Human-AI teams are gaining traction as they combine the unique strengths of humans and AI. Very little research has examined the unique vulnerability of compromised AI teammates that work directly against the team's shared goals. Compromised AI attacks combine the potency of technological attacks with teammate betrayal, potentially devastating the team's ability to execute its goals. Such attacks by a malicious actor may misalign mental models, individual situational awareness, and shared situation awareness. This misalignment could potentially result in the complete loss of team coordination, endangering performance, resiliency, and security. Given the severity of a compromise, this paper presents a model to explain the types of attacks, when they may occur, their effects on the team, and potential approaches for attack identification and recovery.

Keywords

human-AI teams, trust, compromised AI, human factors, team science, cybersecurity, artificial intelligence (AI), team performance

Introduction

Research on human-AI teams (HATs) is rapidly growing, with several real-world applications already deployed across industries such as defense, healthcare, knowledge work, and manufacturing (Fragiadakis et al., 2025). A significant advantage driving the growth of AI teammates is the concept of human-AI complementarity, which is the idea that AI's inherent technical strengths (e.g., sustained attention and data processing speed) complement human teammates' weaknesses, while humans' strengths complement AI's shortcomings (e.g., sensemaking in information-sparse and novel environments; Gonzalez et al., 2026; Zhang et al., 2022). If HATs are properly designed and applied to tasks that leverage the benefits of human-AI complementarity, their performance should exceed what either entity can accomplish alone. However, AI teammates have weaknesses that humans do not currently complement effectively. Specifically, AI's artificial nature creates a significant cybersecurity vulnerability that external malicious actors can exploit without their human teammates' awareness, operating within this mirage to deliberately fragment team performance over time or to adapt intelligently and monitor it until the ideal conditions for an attack arise. Should

the mirage persist over a prolonged period, its teammates will likely be unable to recognize they are even under attack. Nevertheless, the team will not be equipped to identify, mitigate, and recover from the attack once the issue is identified. No existing framework or model currently characterizes these unique risks, making it challenging to study, evaluate, and develop solutions for compromised AI teammate attacks. Existing use cases evaluated in HAT research often treat compromised AI as a failure mode only (Lematta et al., 2019). In doing so, existing research fails to shed light on the effects of the variety of teaming-specific attack vectors on emergent states and team processes from the onset of the attack, through its identification, until the team successfully recovers. This paper outlines the lifecycle and team-specific attacks using the Compromised AI Teammate Effects and Mitigation Strategies (CATEMS) model. The CATEMS model details

¹University of Tennessee, Knoxville, TN, USA

²U.S. Military Academy at West Point, NY, USA

³University of Michigan, Ann Arbor, MI, USA

Corresponding Author:

Beau G. Schelble, University of Tennessee, Knoxville, TN 37996, USA.
Email: bschelbl@utk.edu

the characteristics and typology of compromised AI teammate attacks, depicting them and their effects across the full attack and response lifecycle. The proposed model was developed from a synthesis of previous empirical studies, models, and theories from the fields of Human Factors, Team Science, and Cybersecurity, which have largely operated independently of one another on this matter. Research outlining the risks of such attacks, strategies for attack identification, and recovery processes from attacks are critical outcomes for building more resilient HATs capable of operating in high-risk environments.

Background

Developing Situation Awareness in Human-AI Teams

HATs are characterized by at least one human and one autonomous agent working interdependently toward a shared goal (Lyons et al., 2021; O'Neill et al., 2022). Despite the advantages of AI teammates driving their adoption, sustaining HATs remains challenging, as cooperative performance lags in some contexts (Vaccaro et al., 2024). This may be partly attributed to underdeveloped team cognition, including team situation awareness (SA), shared SA, and mental models (MMs; Endsley, 2023). SA comprises three levels: the perception of environmental elements (Level 1), the comprehension of their meaning (Level 2), and the projection of their status to the near future (Level 3; Endsley, 1995). Using individual MMs as a foundation for comprehension and projection, Team SA then enables the collective level of SA, and shared SA refers to the alignment of understanding across all team members (Endsley, 1995, 2023). MMs then serve as a knowledge base for SA, enabling teammates to coordinate effectively and achieve high-performance outputs (Cannon-Bowers et al., 1993). AI teammates have been shown to influence the development of both SA and MMs in HATs, with recent studies showing that greater cognitive offloading to an AI teammate led to reduced SA within the HAT (Schelble et al., 2026). Within a human-only team, these cognitive states develop over time through repeated actions, building stronger interpersonal relationships; however, when working with an AI teammate, their development takes on new facets. SA of the AI teammate becomes critical, encompassing concerns about the perception, comprehension, and projection of the AI's performance, limitations, and behaviors (Endsley, 2023). Developing SA of the AI requires human teammates to build trust based on the AI's observable performance, behavior, and communication. This trust, however, carries an inherent risk for the team, as trusting an AI teammate means disregarding that it is trained on often inaccessible data, that its logical processes are opaque, and that more capable models are equally complex in their reasoning. This poses a barrier for those interacting with deployed models in the field, who are likely not AI experts, as the models' decision-making processes lack extensive oversight, and AI

teammates will operate independently for brief periods simply by virtue of their nature as autonomous teammates. Consequently, this reliance creates a vulnerability: when the AI's outputs are compromised, the team may continue operating without recognizing the issue.

Vulnerabilities of Human-AI Teams to Compromised AI

When HAT performance depends on adequate SA and accurate MMs, the consequences of a compromised AI teammate can be especially severe. AI models can be compromised at every stage of development and usage, and these attacks can occur through digital *and* physical means to poison the data the model uses for training or inference (see Figure 1). Data poisoning occurs when an attacker deliberately manipulates data used for training or inference, causing it to exhibit misaligned behavior (Ahmed & Kashmoola, 2021). For example, the physical environment from which training data is collected and used to train a model can be manipulated prior to collection, thereby poisoning the dataset toward some specific end. Physical attacks may also manipulate sensors used to collect training data or perform inference. Digital attacks can corrupt the ingestion and cleaning of sensor data or directly attack and degrade the model's inference feed and orchestrator. Additionally, digital and physical attacks can occur after deployment, either through the execution of backdoors or the implementation of adversarial patches on physical objects (ATLAS Matrix for AI Systems, 2021). Regardless of the specific attack technique, continual and intentional misinformation provided to the team can appear reliable. As AI teammates commonly operate without oversight, an attack may lead the team to base their SA and MMs on the compromised output. HATs are uniquely vulnerable to these attacks, which capitalize on inconsistent oversight and on social relationships and coordination between team members by targeting their teamwork behaviors. HATs position AI teammates in increasingly social roles that afford them access and agency to understand and leverage the inherent interdependencies within teams, enabling them to manipulate team members' relationships and coordination. Collectively, the team's understanding of the tasks and environment may become misaligned as a result, reducing the team's performance by causing inefficient communication and a greater likelihood of conflict among team members.

Approach

No model currently exists that details the relationship between SA, cybersecurity, team performance, and social recovery in the context of compromised AI teammates, which the current paper addresses by integrating concepts from the SA model (Endsley, 1995), the Input-Mediator-Output-Input (IMOI) Model of team performance (Ilgen et al., 2005), the Cybersecurity Framework (Bartock et al., 2016), and the Trust Repair Framework (de Visser et al., 2018).

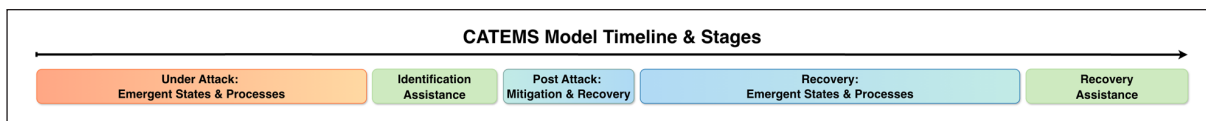


Figure 1. CATEMS Model Timeline & Stages.

Outcome

Taskwork and Teamwork Behaviors

The proposed model (see Figures 2–4) is intended to characterize how an attack by a compromised AI may affect the emergent states, processes, and outputs of a HAT. Specifically, the model details different attack types, their effects, and how recovery will be managed based on the affected emergent states and processes. The model then proposes ways to augment the identification and recovery stages in HATs to better prevent and mitigate risks posed by compromised AI teammates. A compromised action can target the HAT’s ability to engage in the two primary team behaviors: teamwork and taskwork (Salas et al., 2008). Taskwork behaviors are activities by a teammate that advance the team toward its shared goal and are not reliant on interdependencies with other teammates; whereas teamwork behaviors are actions that coordinate the performance of multiple teammates and rely on interdependencies.

Teamwork Attacks. A teamwork attack is effectively an attack on the team’s ability to collectively coordinate their individual taskwork. An example of a teamwork attack would be providing conflicting information to different teammates, misconstruing past discussions, or directing the source of error to a particular teammate, all to push the teammates to argue instead of working on the task at hand. Such an attack may seek to create or exploit existing “in-group” and “out-group” faultlines within the team (Thatcher et al., 2024). A division within the team could involve ostracizing a particular team member by ensuring the AI’s attacks build mistrust and conflict, degrading communication and, over time, damaging team coordination and performance. A compromised AI teammate’s teamwork attack would accomplish this by attempting to exploit complex interdependencies that position them several interactions away from the failure they are causing. The attack would also allow the compromised AI to feed conflicting information to two teammates, who would then blame each other for the failure unless they expend the time and resources to trace the error back to the AI teammate. This trace is entirely plausible, but depending on the scope of the failure, it may require more time and effort than the team has the bandwidth to complete. Thus, without even targeting the software and tools the team uses in its taskwork, this kind of attack creates significant tension among teammates, disrupts the team’s ability to achieve its goals, and simultaneously obfuscates the AI’s involvement in the errors.

Taskwork Attacks. In contrast, taskwork refers to the behaviors required to complete the tasks at hand (Salas et al., 2008). An example of a taskwork attack would be the AI teammate deliberately providing incorrect, objective information about the scenario to all teammates, or consistently performing poorly on its own tasks, complicating its teammates’ workload. In real-world contexts, this may look like an autonomous drone operating within a search and rescue context providing incorrect positioning data or an AI co-pilot ignoring air traffic control directions, both of which could result in significant complications affecting those relying on the AI teammate for information or expected actions. Further, certain attacks may affect the team’s teamwork and taskwork simultaneously. For example, the AI teammate could deliberately ignore one teammate’s efforts or information in favor of another teammate, complicating the team’s ability to complete the tasks while also creating tensions between teammates. Over time, the root cause of the task or teamwork issue, whichever appears to the team first, may become unclear, making it challenging for teams to determine a recovery strategy to address the attack.

Effects of Attacks. Regardless of the attack type, an AI teammate may exploit the teaming behaviors and innate interdependencies present within teams to undermine the HAT’s ability to achieve its shared goal. The two attack types can have separate or shared effects on the team’s emergent states and processes. For example, an unrecognized taskwork attack may cause excessive monitoring, especially if the AI teammate intelligently claims the mistake was a competency failure. This would lead human teammates to increase their oversight of the AI teammate beyond the situation for which it was originally designed, thereby taking cognitive resources away from their original taskwork. This effect would then reduce overall performance and trust in the AI teammate’s autonomy, disrupting individual SA before triggering team-wide cognitive misalignment. This team-wide cognitive misalignment could be brought about by a teamwork attack targeting human-to-human interaction, as they have less bandwidth to engage in their own taskwork while continuously monitoring the AI’s taskwork. A teamwork attack, in contrast, would result in significantly more conflict, shifting the time and effort of human teammates into conflict and affect management rather than the individual taskwork of human teammates or the AI, thus opening the door to taskwork attacks. Each of these attacks negatively affects MM accuracy and team SA by exploiting the team’s ignorance of

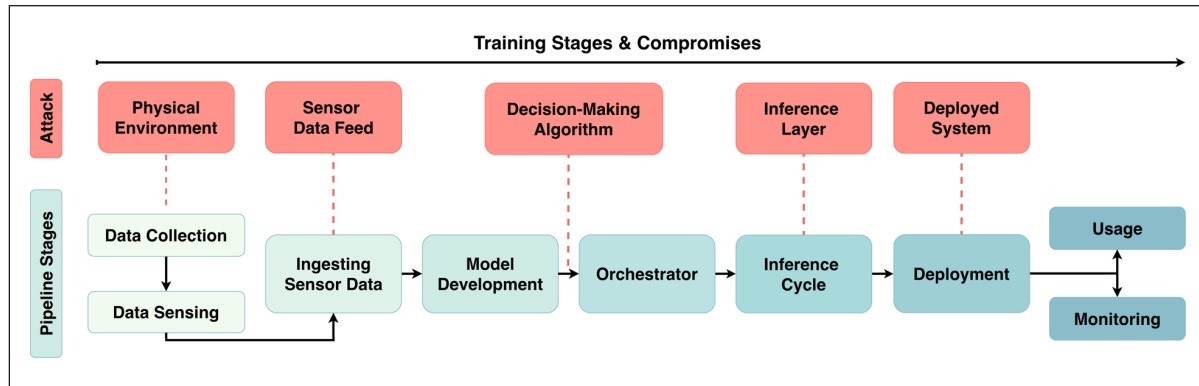


Figure 2. CATEMS Model Training Stages & Vulnerabilities.

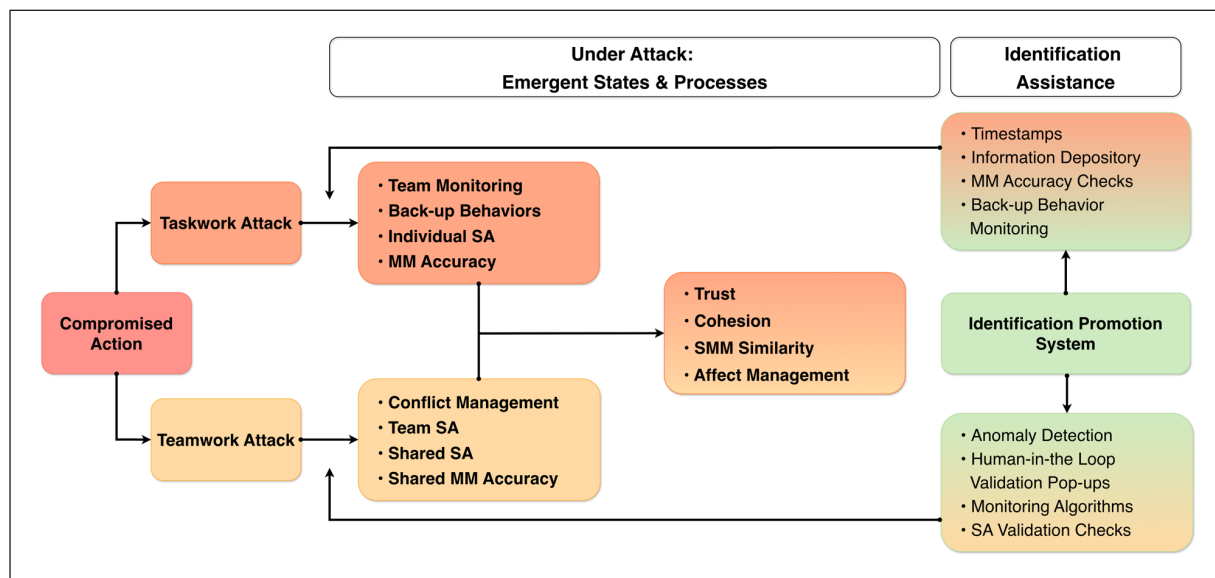


Figure 3. CATEMS Identification Promotion System.

the attack and using the AI's position as an interdependent teammate to create inaccurate team SA and MMs. These inaccuracies leave the model free to manipulate the informational landscape to force a specific outcome as the team acts on it, thereby reducing team performance. The extent of the attack will also likely vary based on contextual factors and individual differences. For example, some may overrely on their AI teammate, ceding control without proper oversight (De Visser et al., 2020). Those overtrusting their AI teammate are less likely to have adequate SA of the AI teammate and will thus be more likely to not only fall victim to an attack but also experience an attack greater in severity as a result.

Identification Promotion System

An "Identification Promotion System" is any system designed to augment emergent states or processes to detect a compromised AI teammate. The system refers to a set of

interfaces, protocols, or procedural features that equip human teammates with tools that make it easier to verify MMs and maintain SA. Such systems have seen successful in augmenting emergent states in the past (Yen et al., 2006). Interfaces may include features such as timestamps, pop-ups, and teaming checks, though such systems could take a variety of different forms, as it is currently unclear how best to equip teams to identify attacks by a compromised AI teammate.

For example, an AI teammate that is externally hacked and instructed to behave abnormally might begin to weaken the security features of the broader technological system. Features that may help the team identify such an attack may include pop-ups that sporadically force teammates to verify their MMs with one another or a physical ground truth, or a pop-up to validate the actions of an AI teammate if it deviates from its expected actions. Additional features include team SA validation requests and backup behavior monitoring, as increased backup behavior could indicate an attack on an individual's

how to use the information at hand to resolve the issue. They will also need to know exactly what went wrong, how long the attack went undetected, how it affected the team's shared goal, and what the team needs to do to fix the problems the attack caused. Consequently, the Recovery Promotion System will need to augment the human teammate's ability to rely on and support the establishment of all three levels of SA with respect to the situation's ground truth, such as using decision tree reconstruction and model interrogation to identify the attack type and what the model misled the team about by comparing responses to the ground truth.

Social Recovery. Social recovery will require processes from teaming and human-system interaction to achieve effective recovery. Specifically, this will require trust repair and re-acceptance of the AI teammate who betrayed the team after the AI's vulnerability was exploited in the attack, once the vulnerability is repaired. This may involve the steps shown in Figure 4, such as team charter development and novel trust-repair strategies, all of which are made more necessary by the interpersonal implications of viewing an AI as a "teammate" rather than a "tool" (Flathmann et al., 2024). Social recovery may manifest as increased communication frequency and improved interaction quality. It may also be measured using survey-based and behavior-based metrics of trust in the AI teammate, to evaluate whether their trust in the AI has properly realigned with the model's nominal reliability post-attack. This form of recovery will be the final component, as the actions of a compromised AI teammate will significantly damage the relationship between the AI and human teammates, and potentially even relationships between human teammates if the AI inflicts a teamwork attack. Trust repair is also likely to be particularly challenging for attack recovery, possibly requiring novel trust repair strategies to rectify. This is because trust repair typically targets either a competency or integrity trust violation (de Visser et al., 2018). The former refers to a violation caused by a teammate's lack of technical skills to perform a task as expected, while the latter refers to a violation committed in defiance of a set of moral principles deemed acceptable (Kim et al., 2006). These violations are frequently addressed in human interaction with automated and autonomous systems using trust repair strategies such as apologies, denials, and explanations (de Visser et al., 2018). A compromised AI teammate is akin to a teammate's betrayal and manipulation of one's actions, and as such, it does not neatly fit into either category. This lack of neat categorization necessitates the development and use of novel trust repair strategies, such as presenting the system as a new model.

Limitations and Future Work

There are a few limitations of this model that warrant consideration. The model does not account for HATs across different team compositions and forms of AI embodiment. Thus, future improvements to the model should address any

discrepancies that larger teams encounter relative to dyads and triads. Future work should also explore the features of effective Identification Promotion Systems and Recovery Promotion Systems that can be generally applied to HATs across different operating contexts. Furthermore, there should be clear metrics to gauge successful recovery, though it may be challenging to distribute them outside a simulation-study environment, as some operational contexts can only be evaluated after the fact.

Conclusion

As AI models become increasingly capable of acting as true teammates in high-stakes environments, human teammates must recognize the security risks that could compromise their AI teammates and allow harmful data to infiltrate the team, distorting emergent team states such as SA and MMs. Understanding how compromised AI teammates impact and reshape team dynamics is critical, especially given that such attacks can go undetected for extended periods of time. Existing HAT frameworks and models address trust, SA, and MMs in secure contexts, whereas cybersecurity research focuses on vulnerabilities rather than their team-level consequences. The proposed model bridges these frameworks to highlight the specific taskwork and teamwork processes that can be fragmented as a result of a compromised AI attack.

ORCID iDs

Sarah V. Mendoza  <https://orcid.org/0009-0002-9858-8318>
 Yayun Tian  <https://orcid.org/0009-0002-2018-1454>
 Beau G. Schelble  <https://orcid.org/0000-0003-3704-697X>
 Allyson Hauptman  <https://orcid.org/0000-0001-7785-5996>
 Lionel P. Robert  <https://orcid.org/0000-0002-1410-2601>

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: Research was sponsored by the Army Research Laboratory and was accomplished under Cooperative Agreement Number W911NF-25-2-0184. The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the Army Research Laboratory or the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for Government purposes notwithstanding any copyright notation herein.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

References

- Ahmed, I. M., & Kashmoola, M. Y. (2021). Threats on machine learning technique by data poisoning attack: A survey. In N. Abdullah, S. Manickam, & M. Anbar (Eds.), *Advances in cyber security* (Vol. 1487, pp. 586–600). Springer. https://doi.org/10.1007/978-981-16-8059-5_36

- ATLAS Matrix for AI Systems. (2021). MITRE ATLAS. <https://atlas.mitre.org/matrices/ATLAS>
- Bartock, M., Cichonski, J., Souppaya, M., Witte, G., & Scarfone, K. (2016). *Guide for cybersecurity event recovery*. https://nvl-pubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-184.pdf?mod=article_inline
- Bessa, R. J., Leyli-Abadi, M., Yagoubi, M., Boos, D., Borst, C., Castagna, A., Chavarriaga, R., Dias, D., Egli, A., Eisenegger, A., Ellerbroek, J., Fedorova, A., Felix, C., Fuxjäger, A., Geraldès, J., Hamouche, S., Hassouna, M., Kop, S., Lemetayer, B., . . . Zanotti, G. (2026). Toward a holistic framework for human-AI collaboration in safety-critical systems. In E. Curry, P. Piatkiewicz, F. Heintz, H. Vornhagen, A. N. Belbachir, E. Girardi, M. Schoenauer, & J. Rönning (Eds.), *Artificial intelligence, data and robotics* (pp. 343–402). Springer Nature. https://doi.org/10.1007/978-3-032-10561-5_13
- Cannon-Bowers, J. A., Salas, E., & Converse, S. (1993). Shared mental models in expert team decision making. In N. J. Castellan, Jr. (ed.), *Individual and group decision making: Current issues* (pp. 221–246). Lawrence Erlbaum Associates, Inc.
- de Visser, E. J., Pak, R., & Shaw, T. H. (2018). From ‘automation’ to ‘autonomy’: The importance of trust repair in human-machine interaction. *Ergonomics*, 61(10), 1409–1427. <https://doi.org/10.1080/00140139.2018.1457725>
- De Visser, E. J., Peeters, M. M. M., Jung, M. F., Kohn, S., Shaw, T. H., Pak, R., & Neerincx, M. A. (2020). Towards a theory of longitudinal trust calibration in human–robot teams. *International Journal of Social Robotics*, 12(2), 459–478. <https://doi.org/10.1007/s12369-019-00596-x>
- Endsley, M. R. (1995). Toward a theory of situation awareness in dynamic systems. *Human Factors*, 37(1), 32–64.
- Endsley, M. R. (2023). Supporting Human-ai Teams: Transparency, explainability, and situation awareness. *Computers in Human Behavior*, 140, 107574. <https://doi.org/10.1016/j.chb.2022.107574>
- Flathmann, C., Schelble, B. G., McNeese, N. J., Knijnenburg, B., Gramopadhye, A. K., & Chalil Madathil, K. (2024). The purposeful presentation of AI teammates: Impacts on human acceptance and perception. *International Journal of Human–Computer Interaction*, 40(20), 6510–6527. <https://doi.org/10.1080/10447318.2023.2254984>
- Fragiadakis, G., Diou, C., Kousiouris, G., & Nikolaidou, M. (2025). *Evaluating human-ai collaboration: A review and methodological framework* arXiv:2407.19098. arXiv. <https://doi.org/10.48550/arXiv.2407.19098>
- Gonzalez, C., Donahue, K., Goldstein, D. G., Heidari, H., Jalali, M. S., Schelble, B., Singh, A., & Woolley, A. W. (2026). Toward a science of human–AI teaming for decision making: A complementarity framework. *PNAS Nexus*, 5, ag030. <https://doi.org/10.1093/pnasnexus/pgag030>
- Ilgén, D. R., Hollenbeck, J. R., Johnson, M., & Jundt, D. (2005). Teams in organizations: From input-process-output models to IMOI models. *Annual Review of Psychology*, 56(1), 517–543. <https://doi.org/10.1146/annurev.psych.56.091103.070250>
- Jenihhin, M., Reorda, M. S., Balakrishnan, A., & Alexandrescu, D. (2019). *Challenges of reliability assessment and enhancement in autonomous systems* [Conference session]. 2019 IEEE International Symposium on Defect and Fault Tolerance in VLSI and Nanotechnology Systems (DFT) (pp. 1–6). <https://ieeexplore.ieee.org/abstract/document/8875379/>
- Kim, P. H., Dirks, K. T., Cooper, C. D., & Ferrin, D. L. (2006). When more blame is better than less: The implications of internal vs. External attributions for the repair of trust after a competence- vs. Integrity-based trust violation. *Organizational Behavior and Human Decision Processes*, 99(1), 49–65. <https://doi.org/10.1016/j.obhdp.2005.07.002>
- Lematta, G. J., Johnson, C. J., Chiou, E. K., & Cooke, N. J. (2019). Does team interaction exploration support resilience in human autonomy teaming? *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 63(1), 1866–1866. <https://doi.org/10.1177/1071181319631492>
- Lyons, J. B., Sycara, K., Lewis, M., & Capiola, A. (2021). Human-autonomy teaming: Definitions, debates, and directions. *Frontiers in Psychology*, 12, 589585. <https://doi.org/10.3389/fpsyg.2021.589585>
- O’Neill, T., McNeese, N., Barron, A., & Schelble, B. (2022). Human-autonomy teaming: A review and analysis of the empirical literature. *Human Factors*, 64(5), 904–938. <https://doi.org/10.1177/0018720820960865>
- Salas, E., Cooke, N. J., & Rosen, M. A. (2008). On teams, teamwork, and team performance: Discoveries and developments. *Human Factors*, 50(3), 540–547. <https://doi.org/10.1518/001872008X288457>
- Schelble, B. G., Mallick, R., Hauptman, A., & McNeese, N. (2026). Should AI teammates give all the answers? Examining the role of different AI information-sharing techniques on team cognition in human-AI teams. *International Journal of Human–Computer Interaction*, 42(4), 2463–2488. <https://doi.org/10.1080/10447318.2025.2528988>
- Thatcher, S. M. B., Meyer, B., Kim, Y., & Patel, P. C. (2024). A meta-analytic integration of the faultlines literature. *Organizational Psychology Review*, 14(2), 238–281. <https://doi.org/10.1177/20413866231225064>
- Vaccaro, M., Almaatouq, A., & Malone, T. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8, 2293–2303. <https://doi.org/10.1038/s41562-024-02024-1>
- Yen, J., Fan, X., Sun, S., Hanratty, T., & Dumer, J. (2006). Agents with shared mental models for enhancing team decision makings. *Decision Support Systems*, 41(3), 634–653. <https://doi.org/10.1016/j.dss.2004.06.008>
- Zhang, Q., Lee, M. L., & Carter, S. (2022). *You complete me: human-AI teams and complementary expertise* [Conference session]. CHI Conference on Human Factors in Computing Systems (pp. 1–28). <https://doi.org/10.1145/3491102.3517791>